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LOCK DEVICE FOR OPENING/CLOSING BODY

Abstract

A lock device for an opening-closing body includes: a base member; and an operating member. The operating member has an operating plate portion, an operating portion, an operating load transmission wall portion protruding, the shaft portions disposed so as to protrude from both sides of the operating load transmission wall portion or the shaft support portions, and a stopper portion connected to the operating load transmission wall portion on a side of the operating portion with respect to the shaft portions or the shaft support portions. The base member has a pair of deformation restriction wall portions disposed outside portions of the operating load transmission wall portion where the shaft portions or the shaft support portions are provided, the shaft support portions or the shaft portions, and a stopper receiving portion that is configured to restrict rotation of the operating member.

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Background/Summary

TECHNICAL FIELD
[0001] The present invention relates to a lock device for an opening-closing body for locking the opening-closing body, which is openably and closably attached to an opening portion of a fixed body, in a closed state.
BACKGROUND ART
[0002] For example, an opening-closing body such as a lid is attached to an opening portion formed in a fixed body such as an automobile glove box so as to be openable and closable. A lock device is provided between the opening portion and the opening-closing body, which locks the opening-closing body when it is closed and can unlock the opening-closing body when it is opened.
[0003] For example, Patent Literature 1 listed below describes a lock device having a base member fixed to an outer member, an operating member rotatably supported by the base member, and a rod that can be engaged with and disengaged from a locked portion of a fixed member in conjunction with rotation of the operating member, in which the operating member has an operating portion exposed on a front side of the outer member and a shaft support portion supported on the base member, and the base member has a hook portion that is hooked onto an attachment edge portion formed on the outer member, and a lock claw portion that is engaged with a lock edge portion formed on a back side of the outer member and that restricts movement of the base member in a direction away from a back surface of the outer member.
[0004] The base member has a pair of wall portions arranged opposite each other, an arc-protrusion-shaped shaft support portion protruding from each wall portion, and a stopper portion provided between the pair of wall portions. The operating member has a long plate-shaped base portion, an operating portion provided at one end of the base portion in a longitudinal direction, a pair of side wall portions erected from longitudinal side edges of the base portion, an arc-groove-shaped shaft portion provided on each side wall portion, a cylindrical portion erected from a back side of the base portion, in which a hole for a key cylinder is formed and into which the key cylinder is inserted, and a rotation stopper portion provided at a tip end of the cylindrical portion in an erection direction.
[0005] The pair of side wall portions of the operating member are placed outside the pair of wall portions of the base member, and the pair of shaft support portions are inserted into the pair of shaft portions, so that the operating member is rotatably attached to the base member.
[0006] When the operating member is rotated to the maximum extent with respect to the base member, the rotation stopper portion of the operating member comes into contact with the stopper portion of the base member, restricting further rotation.
CITATION LIST
Patent Literature
[0007] Patent Literature 1: WO2017/086427A1
SUMMARY OF INVENTION
Technical Problem
[0008] In the case of the lock device of Patent Literature 1, when the operating member is rotated to the maximum extent with respect to the base member, an operating force (operating load) of the operating member is transmitted from the rotation stopper portion through the cylindrical portion in which the hole for the key cylinder is formed and the base portion to the pair of side wall portions

and acts on the pair of shaft portions. In this case, in order to increase breaking strength of the operating member, it is necessary to increase rigidity of the rotation stopper portion, cylindrical portion, base portion, and pair of side wall portions, but this leads to an increase in mass, making it difficult to increase the breaking strength of the operating member, which is inconvenient.

[0009] Therefore, an object of the present invention is to provide a lock device for an opening-closing body that can increase breaking strength of an operating member when the operating member is rotated to the maximum extent with respect to a base member without increasing rigidity of the operating member.

Solution to Problem

[0010] In order to achieve the above object, the present invention provides a lock device for an opening-closing body that is configured to be openably and closably attached to an opening portion of a fixed body, the lock device including a lock portion provided on one of the fixed body or the opening-closing body, a lock member that is disposed on another of the fixed body or the opening-closing body and that is configured to engage with and disengage from the lock portion, a base member that is fixed to the other of the fixed body or the opening-closing body, and an operating member that is rotatably supported on the base member via a rotation portion and that is configured to operate the lock member, in which the rotation portion is formed by shaft portions provided on one of the operating member or the base member, and shaft support portions provided on another of the operating member or the base member to rotatably support the shaft portions, the operating member has an operating plate portion extending in a plate shape, an operating portion provided at one end of the operating plate portion in an extension direction, an operating load transmission wall portion protruding from a back side of the operating plate portion, the shaft portions disposed so as to protrude from both sides of the operating load transmission wall portion toward the outside of the operating member or the shaft support portions disposed on the operating load transmission wall portion, and a stopper portion connected to the operating load transmission wall portion on a side of the operating portion with respect to the shaft portions or the shaft support portions, and the base member has a pair of deformation restriction wall portions disposed outside portions of the operating load transmission wall portion where the shaft portions or the shaft support portions are provided, the shaft support portions disposed on the pair of deformation restriction wall portions or the shaft portions disposed to protrude from the pair of deformation restriction wall portions toward the operating load transmission wall portion, and a stopper receiving portion that is capable of coming into contact with the stopper portion and that is configured to restrict rotation of the operating member.

ADVANTAGEOUS EFFECTS OF INVENTION

[0011] According to the present invention, when the operating portion is operated to rotate the operating member to the maximum extent with respect to the base member, the stopper portion comes into contact with the stopper receiving portion, restricting the rotation of the operating member. In this case, since the stopper portion is connected to the operating load transmission wall portion on which the shaft portion or the shaft support portion is arranged, the operating load received by the stopper portion from the stopper receiving portion is transmitted to the shaft portion via the operating load transmission wall portion, and the portions of the operating load transmission wall portion on which the shaft portions or the shaft support portions are arranged, or the shaft portions arranged to protrude from both sides of the operating load transmission wall portion or the shaft support portions arranged on the operating load transmission wall portion, try to be deformed so as to expand outward from the operating member, but the deformation is restricted by the pair of deformation restriction wall portions. As a result, the breaking strength (resistance to breaking of the operating member) of the operating member can be increased without increasing the rigidity of the operating member.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. **1** is an exploded perspective view illustrating a first embodiment of a lock device for an opening-closing body according to the present invention.

[0013] FIG. **2** is an enlarged exploded perspective view of a main part of the lock device.

[0014] FIG. **3** is a perspective view of an operating member forming the lock device, as viewed from a direction different from that of FIG. **2**.

[0015] FIG. **4** is a rear view of the operating member forming the lock device.

[0016] FIG. **5** is an enlarged perspective view of the lock device.

[0017] FIG. **6** is a cross-sectional view taken along the arrow line E-E in FIG. **5**.

[0018] FIG. **7** is a cross-sectional view taken along the arrow line G-G in FIG. **5**.

[0019] FIG. **8** is a cross-sectional view taken along the arrow line H-H in FIG. **5**.

[0020] FIG. **9** is a cross-sectional view when the operating member is operated with respect to a base member from a state illustrated in FIG. **6**.

[0021] FIG. **10** is an explanatory diagram of a state in which the opening-closing body is closed by the lock device.

[0022] FIG. **11** is an explanatory diagram of a state in which the opening-closing body is opened by the lock device.

[0023] FIG. **12** is an explanatory diagram illustrating a modified example of an operating load transmission wall portion.

[0024] FIG. **13** is a perspective view illustrating a second embodiment of the lock device for the opening-closing body according to the present invention.

[0025] FIG. **14** is a perspective view of an operating member forming the lock device.

[0026] FIG. **15** is a perspective view of a cover member forming the lock device, as viewed from a different direction from that of FIG. **13**.

[0027] FIG. **16** is a cross-sectional view taken along the arrow line I-I of FIG. **13**.

[0028] FIG. **17** is a cross-sectional view taken along the arrow line J-J of FIG. **13**.

[0029] FIG. **18** is an explanatory diagram of the lock device when the operating member is operated with respect to a base member.

[0030] FIG. **19** is a cross-sectional view illustrating a third embodiment of the lock device for the opening-closing body according to the present invention.

[0031] FIG. **20** is a cross-sectional view seen from a different direction from that of FIG. **19**.

[0032] FIG. **21** is a rear view of an operating member forming the lock device.

[0033] FIG. **22** illustrates a fourth embodiment of the lock device for the opening-closing body according to the present invention, illustrating a perspective view of an operating member forming the lock device.

[0034] FIG. **23** is a perspective view of a cover member forming the lock device.

[0035] FIG. **24** is an explanatory diagram illustrating an assembly process of the lock device.

[0036] FIG. **25** is a cross-sectional view of the lock device.

DESCRIPTION OF EMBODIMENTS

First Embodiment of Lock Device for Opening-Closing Body

[0037] Hereinafter, a first embodiment of a lock device for an opening-closing body according to the present invention will be described with reference to the drawings.

[0038] As illustrated in FIG. **1**, a lock device **10** for an opening-closing body (hereinafter referred to as the “lock device **10**”) in this embodiment is used for locking an opening-closing body **2**, such as a glove box, which is openably and closably attached to an opening portion la of a fixed body **1**, such as an instrument panel of a vehicle.

[0039] The lock device **10** of this embodiment is mainly formed by a pair of lock portions **1***b*, **1***b* provided at the opening portion la of the fixed body **1**, a pair of lock members **20**, **20** arranged on the opening-closing body **2** and engaging with and disengaging from the pair of lock portions **1***b*, **1***b*, a rotation member **25** rotatably supported on the opening-closing body **2** and sliding the pair of lock members **20**, **20** in conjunction with each other, urging means (torsion spring **27**) that urges the pair of lock members **20**, **20** in a direction to engage with the pair of lock portions **1***b*, **1***b*, a base member **30** fixed to the opening-closing body **2**, and an operating member **50** that is rotatably supported on the base member **30** via a rotation portion and operates the pair of lock members **20**, **20**. In this embodiment, hole-shaped lock portions **1***b*, **1***b* are provided on both inner surfaces of the opening portion **1***a* of the fixed body **1** in a width direction. In addition, the rotation portion in this embodiment is formed by a shaft portion **61** provided on the operating member **50** and a shaft support portion **37** provided on the base member **30** for supporting the shaft portion **61** rotatably.

[0040] As illustrated in FIGS. **2** and **3**, the operating member **50** includes an operating plate portion **51** extending in a plate shape with a predetermined length, an operating portion **52** provided at one end of the operating plate portion **51** in an extension direction, an operating load transmission wall portion **57** (hereinafter also simply referred to as a "transmission wall portion **57**") protruding from a back side of the operating plate portion **51**, the above-mentioned shaft portions **61**, **61** arranged to protrude from both sides of the transmission wall portion **57** toward the outside of the operating member **50**, and a stopper portion **63** connected to the transmission wall portion **57** on the operating portion **52** side with respect to the shaft portion **61**. In the present invention, the stopper portion **63** being connected to the transmission wall portion **57** means that the stopper portion **63** is directly connected to the transmission wall portion **57** and formed integrally therewith.

[0041] As illustrated in FIGS. **2**, **5**, and the like, the base member **30** includes a pair of deformation restriction wall portions **35**, **35** (hereinafter also simply referred to as a "restriction wall portion **35**") arranged outside a part of the transmission wall portion **57** where the shaft portion **61** is provided, the above-mentioned shaft support portions **37**, **37** formed on the pair of restriction wall portions **35**, **35** and rotatably supporting the shaft portions **61**, **61**, and a stopper receiving portion **41** that is capable of coming into contact with the stopper portion **63** and thereby restricts rotation of the operating member **50**.

[0042] As illustrated in FIG. **1**, in this embodiment, hole-shaped lock portions **1***b*, **1***b* are provided on both inner surfaces of the opening portion **1***a* of the fixed body **1** in the width direction. Also as illustrated in FIG. **1**, the opening-closing body **2** in this embodiment is formed by an outer member **3** arranged on an inner side of an vehicle interior and an inner member **4** arranged on a back side of the outer member **3**. An accommodation recess portion **5** having a horizontally elongated rectangular concave shape is formed on an upper side of one side in the width direction of the outer member **3**. The base member **30** and the operating member **50** are accommodated in this accommodation recess portion **5** (see FIG. **6**). Furthermore, a mounting hole **6** having a long hole shape is formed in a bottom portion **5***a* of the accommodation recess portion **5**.

[0043] The base member **30** is attached from a front side of the mounting hole **6**. The "front side" or "front surface side" here refers to the side or surface located in a direction in which the opening-closing body opens from the opening portion of the fixed body such as a vehicle. When the fixed body is provided on a vehicle, the "front side" or "front surface side" may refer to a vehicle inner space side. The "back side" or "back surface side" refers to the side opposite the "front side" or "front surface side", that is, the side or surface located in a direction in which the opening-closing body closes. The terms "front side", "front surface side", "back side", and "back surface side" have the same meanings not only for the mounting hole, but also for other members (such as the base member **30** and the operating member **50**) described below.

[0044] As illustrated in FIG. **1**, the inner member **4** is generally box-shaped with an open top. An upper portion of the front surface side (front side) of the inner member **4** that faces the outer member **3** is provided with a lock arrangement recess portion **4***a* having a horizontally elongated concave shape in which the pair of lock members **20**, **20** are slidably arranged. Lock insertion holes

4*b*, **4***b* are formed on both sides of the lock arrangement recess portion **4***a* in a longitudinal direction. A rotation member support portion **7** for supporting the rotation member **25** in a rotatable manner protrudes from the front surface side of the bottom surface of the lock arrangement recess portion **4***a*.

[0045] As illustrated in FIGS. **1** and **10**, the pair of lock members **20**, **20** in this embodiment have the same shape, with a base end **21** bent in a crank shape and extending linearly toward a tip end side, and a tip end **22** engages with and disengages from the lock portion **1***b* (see FIGS. **10** and **11**). A frame portion **23** having a quadrangular frame shape is provided on the tip end **22** side of each lock member **20**.

[0046] The rotation member **25** has a substantially diamond shape with a long axis (see FIG. **10**), and its longitudinal intermediate portion is rotatably supported by the rotation member support portion **7**, and both longitudinal ends are respectively connected to the base ends **21**, **21** of the pair of lock members **20**, **20**. When the rotation member **25** rotates in a predetermined direction, the pair of lock members **20**, **20** slide synchronously through the rotation member **25**. The torsion spring **27** is attached to the rotation member **25**, and the rotation member **25** is urged to rotate as illustrated in a direction of the arrow A in FIG. **10**, whereby the tip ends **22**, **22** of the pair of lock members **20**, **20** are urged in a direction (a direction of the arrow B in FIG. **10**) to engage with the pair of lock portions **1***b*, **1***b*. The tip ends **22**, **22** of the pair of lock members **20**, **20** are inserted through lock insertion holes **4***b*, **4***b* provided in the lock arrangement recess portion **4***a*.

[0047] Next, the base member **30** will be described in detail with reference to FIG. **2** and FIGS. **5** to **8** and the like.

[0048] The base member **30** in this embodiment is formed by a bottom wall **31** having a long plate shape that is long in one direction, a pair of side wall portions **32**, **32** that are erected from both side edges along the long sides of the bottom wall **31** and extend along an extension direction of the bottom wall **31**, a connecting wall **33** that is arranged at one end side of the bottom wall **31** in the longitudinal direction and that connects the pair of side wall portions **32**, **32**, and a connecting wall **34** that is arranged at another end side of the bottom wall **31** in the longitudinal direction and that connects the pair of side wall portions **32**, **32**, and has a substantially long box shape that is open on a ceiling (upper) side opposite the bottom wall **31**.

[0049] In the following description, the direction in which the bottom wall **31** extends is referred to as an “extension direction X**1**”, and the direction orthogonal to this extension direction X**1** is referred to as a “width direction Y**1**” (the same applies to the base member **30** itself and to each part other than the bottom wall **31** that forms the base member **30**).

[0050] A slide groove **31***a* is formed in the bottom wall **31** to slidably receive an operating lever **65** (see FIG. **3**) of the operating member **50**. Also, from the upper side (an opposite side to the bottom wall **31**) of the pair of side wall portions **32**, **32** and a position near the connecting wall **34**, protruding pieces **35***a*, **35***a* having a substantially trapezoidal mountain shape are provided.

[0051] As illustrated in FIG. **1**, the portion of the pair of side wall portions **32**, **32** near another end in the extension direction X**2** including the protruding pieces **35***a* forms a pair of restriction wall portions **35**, **35**. The pair of restriction wall portions **35**, **35** are disposed to face the outside of the portion (here, the portion of the transmission wall portion **57** where the shaft portions **61**, **61** are provided on a pair of flat surfaces **57***b*, **57***b* described below) of the transmission wall portion **57** where the shaft portion **61** is provided, and serve as a portion that restricts deformation of the transmission wall portion **57**. Inner surfaces (opposing surfaces) of the pair of restriction wall portions **35**, **35** are formed to be parallel to each other.

[0052] In addition, the shaft support portion **37** having a circular hole shape is formed penetrating the protruding pieces **35***a*, **35***a* forming the pair of restriction wall portions **35**, **35**. Then, as illustrated in FIGS. **7** and **8**, each shaft portion **61** of the operating member **50** is inserted from the inside of each shaft support portion **37**, and the operating member **50** is rotatably supported by the base member **30**.

[0053] In addition, on an outer surface of each side wall portion **32**, on the bottom wall **31** side, there is provided a flexible elastic engagement piece **38** formed through a slit **38***a* having a U shape, and a rib **39** protruding from a position adjacent to the elastic engagement piece **38**. The elastic engagement piece **38** engages with a rear peripheral edge of the mounting hole **6**, and the rib **39** engages with a front peripheral edge of the mounting hole **6** (see FIG. **7**).

[0054] A pair of hooks **40**, **40** that engage with the rear peripheral edge of the mounting hole **6** protrude from both sides of the connecting wall **33** in the width direction Y**1** on the bottom wall **31** side. Furthermore, a notch **41***a* is formed in an intermediate portion of the connecting wall **33** in the width direction Y**1** from the bottom wall **31** toward the ceiling opening, and the stopper receiving portion **41** is provided on the connecting wall **33** on an opposite side to the bottom wall **31** side through the notch **41***a*. As illustrated in FIG. **9**, when the operating portion **52** of the operating member **50** is rotated to the maximum extent with respect to the base member **30** so as to move away from the base member **30** and the bottom portion **5***a* of the accommodation recess portion **5**, the stopper receiving portion **41** comes into contact with the stopper portion **63**, and further rotation of the operating member **50** is restricted.

[0055] A pair of elastic engagement claws **42**, **42** are provided on both sides of the connecting wall **34** in the width direction Y**1** on the bottom wall **31** side, which engage with the rear peripheral edge of the mounting hole **6**. Furthermore, a damper **43** made of rubber is attached to a corner of one end of the base member **30** in the extension direction X**1**, so as to reduce hitting noise when the operating portion **52** of the operating member **50** rotates in a direction approaching the base member **30** or the bottom portion **5***a* of the accommodation recess portion **5**. A coil spring **44** is arranged at a corner of another end of the base member **30** in the extension direction X**1**.

[0056] Next, the operating member **50** will be described in detail with reference to FIGS. **2** to **8** and the like.

[0057] In this embodiment, the operating member **50** is elongated in one direction overall, and includes the operating plate portion **51** having a substantially elongated plate shape that is plate-like and extends with a predetermined length. In the following description, a direction in which the operating plate portion **51** extends is referred to as an “extension direction X**2**”, and a direction orthogonal to the extension direction X**2** is referred to as a “width direction Y**2**” (this applies to the operating member **50** itself, each part other than the operating plate portion **51** that forms the operating member **50**, and also to a cover member **70** described below).

[0058] The operating portion **52** is provided at one end (tip end) of the operating plate portion **51** in the extension direction X**2**, which is a portion that an operator grasps and operates when rotating the operating member **50**. Another end of the operating plate portion **51** in the extension direction X**2** is a base end **53** that is rotatably supported by the pair of shaft support portions **37**, **37** of the base member **30**. Furthermore, a cylinder insertion hole **51***a* having a circular hole shape is formed at the other end side of the operating plate portion **51** in the extension direction X**2**, through which a key cylinder **67** is inserted. A damper abutment portion **51***b* having a substantially cylindrical protrusion shape that abuts against the damper **43** is provided on the back side (the side facing the base member **30**) of the operating plate portion **51**, near one end in the extension direction X**2**.

[0059] Furthermore, a pair of side walls **55**, **55** extending along the extension direction X**2** of the operating plate portion **51** are erected from both side edges of the operating plate portion **51** in the width direction Y**2** toward the back side (toward the base member **30**). Furthermore, recess portions **59**, **59** are formed on another end side of the pair of side walls **55**, **55** in the extension direction X**2** to receive a pair of protruding pieces **35***a*, **35***a* of the base member **30**.

[0060] The transmission wall portion **57** having a substantially cylindrical shape into which the key cylinder **67** can be inserted is erected on the rear side of the operating plate portion **51** from a rear peripheral edge of the cylinder insertion hole **51***a*. As will be described in detail below, as illustrated in FIG. **8**, this transmission wall portion **57** transmits an operating load input via the stopper portion **63** to the pair of shaft portions **61**, **61** (see the arrow F**2**′ in FIG. **8**), and is capable of bending and deforming radially outward and radially inward. The operating load is mainly transmitted through a semicircular portion **57***a* (see FIG. **8**) of the transmission wall portion **57** that is located on the operating portion **52** side with respect to the pair of shaft portions **61**, **61**.

[0061] In addition, the flat surfaces **57***b*, **57***b* that are parallel to each other and have a flat surface shape are formed on an outer periphery of both circumferentially opposing sides (both sides in the width direction Y**2**) of the transmission wall portion **57** at positions that align with the pair of recess portions **59**, **59**. A pair of shaft portions **61**, **61** protrude coaxially from a position (a position on a base end side of the transmission wall portion **57** in an erection direction) of the pair of flat surfaces **57***b*, **57***b* on the operating plate portion **51** side.

[0062] Furthermore, as illustrated in FIG. **4**, when the operating member **50** is viewed from the back side (when viewed from the rear side), the pair of shaft portions **61**, **61** extend from a position near the other end (near the base end **53**) of the operating member **50** in the extension direction X**2** with respect to an axis C of the transmission wall portion **57** along the width direction Y**2** of the operating member **50**, and are connected to the transmission wall portion **57**. Also, as illustrated in FIG. **3**, at the tip end of each shaft portion **61**, a tapered surface **61***a* is formed that is cut obliquely from the base end side in an erection direction of the transmission wall portion **57** toward the tip end side in the erection direction, making it easy to insert the shaft portion **61** from the inside of the shaft support portion **37**.

[0063] When assembling the operating member **50** to the base member **30**, the pair of shaft portions **61**, **61** of the operating member **50** are inserted between the pair of protruding pieces **35***a*, **35***a* of the base member **30**, and the operating member **50** is pressed against the base member **30**. The pair of shaft portions **61**, **61** are then inserted into the pair of shaft support portions **37**, **37** of the base member **30** from the inside (see FIGS. **7** and **8**), so that the operating member **50** is rotatably supported against the base member **30**. As a result, the pair of restriction wall portions **35**, **35** of the base member **30** are arranged so as to face the outer sides of the pair of flat surfaces **57***b*, **57***b* (see FIGS. **7** and **8**). In particular, the protruding pieces **35***a*, **35***a* forming the pair of restriction wall portions **35**, **35** are disposed outside the portion of the pair of flat surfaces **57***b*, **57***b* where the pair of shaft portions **61**, **61** are provided. Both the pair of flat surfaces **57***b*, **57***b* and inner surfaces of the pair of restriction wall portions **35**, **35** including the protruding pieces **35***a*, **35***a* have flat surface shape, so that the flat surfaces **57***b* and the inner surface of the restriction wall portion **35** are disposed to face each other in close proximity, and a gap between the flat surface **57***b* and the inner surface of the restriction wall portion **35** is small. As illustrated in FIGS. **6** and **9**, the operating member **50** rotates with respect to the base member **30**, using the pair of shaft portions **61**, **61** as rotation fulcrums, so that the operating portion **52** of the operating plate portion **51** of the operating member **50** approaches and moves away from the base member **30** and the bottom portion **5***a* of the accommodation recess portion **5**.

[0064] As illustrated in FIG. **6**, the coil spring **44** is interposed in a compressed state between the other end of the operating member **50** in the extension direction X**2** and the other end of the base member **30** in the extension direction X**1**. Therefore, as illustrated in FIG. **9**, when the operator rotates the operating member **50** with respect to the base member **30** so that the operating portion **52** of the operating plate portion **51** of the operating member **50** moves away from the base member **30** and the bottom portion **5***a* of the accommodation recess portion **5**, and then releases his or her hand from the operating portion **52**, the operating portion **52** is urged to rotate in a direction approaching the base member **30** and the bottom portion **5***a* of the accommodation recess portion **5**.

[0065] As illustrated in FIG. **3**, the stopper portion **63** protrudes from an outer surface of a tip end of the transmission wall portion **57** in the erection direction on the operating portion **52** side with respect to the pair of shaft portions **61**, **61**. The stopper portion **63** is disposed perpendicular to the pair of shaft portions **61**, **61**, and is connected to the transmission wall portion **57**. That is, the transmission wall portion **57**, the pair of shaft portions **61**, **61**, and the stopper portion **63** are integrally formed continuously. The stopper portion **63** comes into contact with the stopper receiving portion **41** when the operating member **50** is rotated to the maximum extent with respect to the base member **30** so that the operating portion **52** of the operating member **50** is in a direction away from the base member **30** and the bottom portion **5***a* of the accommodation recess portion **5**, as illustrated in FIG. **9**, thereby restricting further rotation of the operating member **50**.

[0066] As illustrated in FIG. **3**, the operating lever **65** is erected from the back side of one end of the operating plate portion **51** in the extension direction X**2**. A part of the operating lever **65** is connected to a portion of the transmission wall portion **57** in the circumferential direction opposite the portion where the stopper portion **63** is provided, and is erected higher than the tip end of the transmission wall portion **57** in the erection direction. The operating lever **65** is inserted from the back side of the slide groove **31***a* of the base member **30** and inserted into the frame portion **23** of one of the lock members **20**. When the operating portion **52** of the operating member **50** rotates in a direction away from the base member **30** or the bottom portion **5***a* of the accommodation recess portion **5**, the operating lever **65** presses the frame portion **23** of one of the lock members **20**, thereby sliding the tip ends **22**, **22** of the pair of lock members **20**, **20** via the rotation member **25** in a direction not to engage with the pair of lock portions **1***b*, **1***b* of the fixed body **1** (see FIG. **11**).

[0067] In this lock device **10**, when the operating portion **52** is operated to rotate the operating member **50** with respect to the base member **30** and the stopper portion **63** comes into contact with the stopper receiving portion **41**, the operating load from the stopper receiving portion **41** is input to the stopper portion **63**, and the operating load is transmitted from the stopper portion **63** to the transmission wall portion **57**, and further transmitted to the shaft portion **61** via the transmission wall portion **57**.

[0068] That is, from the state illustrated in FIG. **6**, when the operating portion **52** of the operating member **50** is grasped and the operating portion **52** is rotated to the maximum extent with respect to the base member **30** in a direction away from the base member **30** and the bottom portion **5***a* of the accommodation recess portion **5** as illustrated by the arrow F**1**, the stopper portion **63** comes into contact with the stopper receiving portion **41** as illustrated in FIG. **9**, restricting further rotation of the operating member **50**. Then, the stopper portion **63** receives an operating load F**1**′ from the stopper receiving portion **41**. That is, the operating load F**1**′ is input from the stopper receiving portion **41** to the stopper portion **63**, and the operating load F**1**′ is transmitted from the stopper portion **63** to the transmission wall portion **57**.

[0069] Then, the operating load F**1**′ is transmitted from the portion of the transmission wall portion **57** where the stopper portion **63** is located, that is, a circumferential center of the semicircular portion **57***a* of the transmission wall portion **57** on the operating portion **52** side with respect to the pair of shaft portions **61**, **61**, to the base end side in an axial direction of the semicircular portion **57***a* as illustrated by the arrow F**2** in FIG. **9**, and is also transmitted to both ends of the semicircular portion **57***a* in the circumferential direction as illustrated by the arrow F**2**′ in FIG. **8**. As a result, the operating load F**1**′ is applied to the portion of the transmission wall portion **57** where the shaft portions **61**, **61** are provided, that is, the portion of the pair of flat surfaces **57***b*, **57***b* forming the transmission wall portion **57** where the pair of shaft portions **61**, **61** are provided. As a result, as illustrated by the arrow F**3** in FIG. **8**, the pair of flat surfaces **57***b*, **57***b* attempt to bend and deform in a direction away from each other (it can also be said that the pair of flat surfaces **57***b*, **57***b* attempt to bend and deform radially outward of the transmission wall portion **57** so that a distance between the base ends of the pair of shaft portions **61**, **61** increases, and further, it can be said that the pair of shaft portions **61**, **61** bend and deform radially outward of the transmission wall portion **57**).

[0070] In this case, the pair of restriction wall portions **35**, **35** are arranged so as to face the pair of flat surfaces **57***b*, **57***b* on the outside of the pair of flat surfaces **57***b*, **57***b*. In particular, the pair of protruding pieces **35***a*, **35***a* forming the pair of restriction wall portions **35**, **35** are arranged so as to face the pair of flat surfaces **57***b*, **57***b* on the outside of the portion of the pair of flat surfaces **57***b*, **57***b* where the pair of shaft portions **61**, **61** are provided. Therefore, the radial outward deflection deformation of the transmission wall portion **57** of the portion of the pair of flat surfaces **57***b*, **57***b* where the pair of shaft portions **61**, **61** are provided and the pair of shaft portions **61**, **61** is restricted. In this embodiment, the key cylinder **67** is inserted and arranged inside the transmission wall portion **57** to restrict the radial inward deflection deformation of the transmission wall portion **57**.

Modified Example

[0071] The shapes, structures, layouts, and the like of the lock portion, lock member, base member, and operating member that form the lock device for the opening-closing body in the present invention, as well as the various parts (the operating plate portion, operating load transmission wall portion, shaft portion, stopper portion, and the like) that form the operating member, and the various parts (the deformation restriction wall portion, shaft support portion, stopper receiving portion, and the like) that form the base member are not limited to the above-described embodiment.

[0072] Furthermore, as described above, this embodiment is applied to, for example, a structure in which a box-shaped glove box is rotatably attached to an opening portion of an instrument panel (in this case, the instrument panel forms the "fixed body" and the glove box forms the "opening-closing body"), but it may also be applied to a structure in which a lid is attached to an opening portion of the instrument panel so that it can be opened and closed (in this case, the instrument panel forms the "fixed body" and the lid forms the "opening-closing body"), and can be widely used for various opening-closing bodies that open and close the opening portion of a fixed body.

[0073] Furthermore, in this embodiment, the mounting hole **6** is formed in the opening-closing body **2** and the lock member **20** is slidably arranged on the opening-closing body **2**, but the mounting hole may be formed in the fixed body and the lock member may be slidably arranged on the fixed body side.

[0074] In addition, although the lock portion **1***b* in this embodiment is hole-shaped, the lock portion may not be hole-shaped, but may be concave, protrusion-shaped, frame-shaped, or the like, and the lock portion may be provided in the opening-closing body rather than in the opening portion of the fixed body.

[0075] Furthermore, the torsion spring **27**, which is the urging means in this embodiment, urges the rotation member **25** to rotate, thereby urging the lock member **20** connected to the rotation member **25** in a direction to engage with the lock portion **1***b* (see FIG. **10**). However, the urging means that urges the lock member in a direction such that the lock member engages with the lock portion may be, for example, a tension spring that pulls one of the lock members toward the lock portion.

[0076] In addition, in this embodiment, as described above, when the operating member **50** is rotated, one of the lock members **20** is slid, and another lock member **20** is slid via the rotation member **25**. However, it may be configured so that the rotation member is rotated by rotating the operating member, thereby sliding the pair of lock members.

[0077] Furthermore, although the opening-closing body **2** in this embodiment is formed by the outer member **3** and the inner member **4**, the opening-closing body may be formed by a single plate material.

[0078] In addition, the shaft support portion **37** formed in the base member **30** has a circular hole shape, but the shaft support portion may be, for example, an arc-shaped groove (this will be described in the embodiment described below) or an arc-shaped protrusion.

[0079] Furthermore, in this embodiment, the pair of shaft portions **61**, **61** are connected to the transmission wall portion **57**, but the shaft portion may be separate from the operating load transmission wall portion (this will be described in the embodiment described below). The shaft portion preferably has a shape that matches the shape of the shaft support portion. For example, when the shaft support portion is an arc-shaped groove as described in paragraph 0053, the shaft portion is preferably an arc-shaped protrusion.

[0080] In addition, in this embodiment, the key cylinder **67** is inserted and positioned inside the transmission wall portion **57** to restrict radially inward deflection deformation of the transmission wall portion **57**, but a configuration in which a key cylinder is not inserted into the operating load transmission wall portion, or a configuration in which a separate member other than the key cylinder is inserted and positioned into the operating load transmission wall portion (this will be described in the embodiment described below), may also be used.

[0081] Furthermore, in this embodiment, a rotation portion that supports the operating member **50** to rotate on the base member **30** is formed by the shaft portion **61** provided on the operating member **50** and the shaft support portion **37** provided on the base member **30**, but the rotation portion may also be formed by a shaft portion provided on the base member and a shaft support portion provided on the operating member

[0082] In this case, the shaft support portion is disposed in the operating load transmission wall portion forming the operating member, and the stopper portion forming the operating member is connected to the operating load transmission wall portion on the operating portion side with respect to the shaft support portion. The shaft portions are disposed so as to protrude from the pair of deformation restriction wall portions forming the base member toward the operating load transmission wall portion. In other words, the shaft portions are disposed to protrude from the pair of deformation restriction wall portions toward the inside of the base member so that the shaft portions protrude toward the operating load transmission wall portion that is disposed inside the pair of deformation restriction wall portions.

[0083] Further, although the transmission wall portion **57** in this embodiment is substantially cylindrical, the operating load transmission wall portion may be, for example, semi-cylindrical, triangular tubular, V-frame-shaped, U-frame-shaped, or the like.

[0084] FIG. **12** illustrates a modified example of an operating load transmission wall portion **68** (hereinafter also simply referred to as a “transmission wall portion **68**”). This transmission wall portion **68** is frame-shaped and includes a pair of inclined ribs **68***a*, **68***a* extending from one end side to another end side of an operating member **50**A in the extension direction X**2** so as to expand obliquely outward, bent ribs **68***b*, **68***b* formed at tip ends of the pair of inclined ribs **68***a*, **68***a* in the extension direction and extending along the extension direction X**2** of the operating member **50**A, a first connecting rib **68***c* connecting the bent ribs **68***b*, **68***b* and extending along the width direction Y**2** of the operating member **50**A, and a second connecting rib **68***d* extending from a base end of the pair of inclined ribs **68***a*, **68***a* in the extension direction along the extension direction X**2** of the operating member **50**A and connected to the center of the first connecting rib **68***c* in the extension direction. Further, a stopper portion **63** is provided at the base ends of the pair of inclined ribs **68***a*, **68***a* and the second connecting rib **68***d* in the extension direction, and shaft portions **61**, **61** protrudes from outer surfaces of the bent ribs **68***b*, **68***b*.

[0085] In the case of such a transmission wall portion **68**, when the operating load input from the stopper receiving portion **41** to the stopper portion **63** is transmitted from the stopper portion **63** to the transmission wall portion **57**, the operating load is transmitted through the pair of inclined ribs **68***a*, **68***a* (see the arrow F**2**′ in FIG. **12**) and acts on the pair of bent ribs **68***b*, **68***b*, so that the pair of bent ribs **68***b*, **68***b* are bent and deformed so as to expand in a direction away from each other (see the arrow F**3** in FIG. **12**). However, even in this case, a pair of deformation restriction wall portions (not illustrated) are arranged on the outside of the pair of bent ribs **68***b*, **68***b* to restrict the deflection deformation of the pair of bent ribs **68***b*, **68***b* in a radial outward direction.

Operational Effects

[0086] Next, operational effects of the lock device **10** having the above configuration will be described.

[0087] In this lock device **10**, the opening-closing body **2** is closed from an state where the opening-closing body **2** is opened from the opening portion **1***a* of the fixed body **1**, and the tip ends **22**, **22** of the pair of lock members **20**, **20** engage with the lock portions **1***b*, **1***b* of the fixed body **1** (see FIG. **10**), thereby locking the opening-closing body **2** in a closed state.

[0088] From this state, when the operating portion **52** of the operating member **50** is rotated in a direction away from the base member **30** and the bottom portion Sa of the accommodation recess portion **5**, as illustrated in FIG. **9**, the tip end **22** of one lock member **20** is retracted in a direction not to engage from the lock portion **1***b* against a rotational urging force of the rotation member **25**, as illustrated in FIG. **11**. In conjunction with this, the tip end **22** of the other lock member **20** is retracted in a direction not to engage from the lock portion **1***b* via the rotation member **25**, so that a locked state of the opening-closing body **2** is released and the opening-closing body **2** can be opened from the opening portion **1***a* of the fixed body **1**.
[0089] In this lock device **10**, as described in paragraph 0043 above, when the operating portion **52** of the operating member **50** is grasped and the operating portion **52** is rotated to the maximum extent with respect to the base member **30**, the stopper portion **63** comes into contact with the stopper receiving portion **41**, and the operating load F**1**′ is input from the stopper receiving portion **41** to the stopper portion **63**, and this operating load F**1**′ is transmitted from the stopper portion **63** to the transmission wall portion **57** (see FIG. **9**). Then, as described in paragraph 0044 above, the operating load F**1**′ is applied to the pair of flat surfaces **57***b*, **57***b* through the semicircular portion **57***a* of the transmission wall portion **57** (see the arrow F**2**′ in FIG. **8**), and as illustrated by the arrow F**3** in FIG. **8**, the portions of the pair of flat surfaces **57***b*, **57***b* where the pair of shaft portions **61**, **61** are provided and the pair of shaft portions **61**, **61** attempt to bend and deform so as to expand in a direction away from each other. In this case, the pair of restriction wall portions **35**, **35** are arranged to face the pair of flat surfaces **57***b*, **57***b* on the outside of the pair of flat surfaces **57***b*, **57***b* (here, the pair of protruding pieces **35***a*, **35***a* are arranged to face them), so that the portions of the pair of flat surfaces **57***b*, **57***b* where the pair of shaft portions **61**, **61** are provided, and the pair of shaft portions **61**, **61** can be restricted from being bent and deformed radially outward of the transmission wall portion **57**. As a result, the breaking strength (resistance to breaking of the operating member **50**) of the operating member **50** can be increased without increasing the rigidity of the operating member **50**.
[0090] In this embodiment, the transmission wall portion **57** is cylindrical shape so that the key cylinder **67** can be inserted therein.
[0091] According to this embodiment, before the key cylinder **67** is inserted into the transmission wall portion **57**, the transmission wall portion **57** can be easily deformed radially inward. Therefore, when the transmission wall portion **57** is disposed between the pair of restriction wall portions **35**, **35** of the base member **30** in order to assemble the operating member **50** to the base member **30**, the transmission wall portion **57** is disposed so that the shaft portion **61** can be easily supported by the shaft support portion **37**, thereby improving the workability of assembling the operating member **50** to the base member **30**. In addition, when the key cylinder **67** is inserted into the transmission wall portion **57** in a state in which the shaft portion **61** is supported by the shaft support portion **37**, the radially inward deformation of the transmission wall portion **57** can be restricted, so that the rigidity of the transmission wall portion **57** itself and the shaft portion **61** and stopper portion **63** provided on the transmission wall portion **57** can be increased.
Second Embodiment of Lock Device for Opening-Closing Body
[0092] FIGS. **13** to **18** illustrate a second embodiment of the lock device for the opening-closing body according to the present invention. Parts that are substantially the same as those in the previous embodiment are given the same reference signs, and their descriptions are omitted.
[0093] A lock device **10**B for the opening-closing body (hereinafter simply referred to as a “lock device **10**B”) of this embodiment has a pair of notches **59**B, **59**B formed on both sides of an operating plate portion **51** along the extension direction X**2**, into which a pair of restriction wall portions **35**, **35** fit, an operating member **50**B further has a cover member **70** arranged on the front side of the operating plate portion **51**, and the cover member **70** is provided with wall portions **75**, **75** that cover the pair of notches **59**B, **59**B.
[0094] As illustrated in FIG. **14**, the operating member **50**B is formed with the pair of notches **59**B,

59B extending along the extension direction X**2** of the operating plate portion **51** on both side edges in the width direction Y**2** on the base end **53** side of the operating plate portion **51**. The pair of notches **59**B, **59**B are arranged so that a pair of protruding pieces **35***a*, **35***a* of the base member **30** can fit into them (see FIG. **17**). A pair of shaft portions **61**, **61** protruding from a pair of flat surfaces **57***b*, **57***b* of a transmission wall portion **57** are arc-shaped protrusions. A pair of shaft support portions **37**, **37** provided on the base member **30** side are arc-shaped grooves.

[0095] As illustrated in FIG. **15**, the cover member **70** has a cover plate portion **71** having substantially elongated plate shape that is plate-like and extends in a predetermined length so as to fit the operating plate portion **51**. A pair of wall portions **75**, **75** extending along the extension direction X**2** of the cover plate portion **71** are erected from both side edges of the cover plate portion **71** in the width direction toward the back side (toward the operating member **50**B side). In addition, a pair of lock protrusions **73***a*, **73***a* are formed on the base end **73** side of the cover plate portion **71** in the extension direction X**2**, which lock onto both sides of the base end **53** of the operating plate portion **51** in the width direction. A tip end side of the cover plate portion **71** in the extension direction X**2** forms an operating portion **72**. Furthermore, a lock frame portion **72***a* into which a tip end **52**B of the operating plate portion **51** fits and locks (see FIG. **16**) is provided in the cover plate portion **71** on the base end **73** side with respect to the operating portion **72**.

[0096] Then, the tip end **52**B of the operating plate portion **51** is inserted into and engaged with the lock frame portion **72***a* of the cover plate portion **71**, and the lock protrusions **73***a*, **73***a* of the cover plate portion **71** are respectively engaged with both sides of the base end **53** of the operating plate portion **51** in the width direction, thereby attaching the cover member **70** to the front side of the operating plate portion **51**. As a result, as illustrated in FIG. **13**, the pair of notches **59**B, **59**B formed in the operating plate portion **51** are respectively covered with the pair of wall portions **75**, **75** of the cover member **70**. As illustrated in FIG. **17**, the pair of protruding pieces **35***a*, **35***a* of the base member **30** are respectively fitted to the pair of notches **59**B, **59**B of the operating member **50**B, and further, the pair of wall portions **75**, **75** of the cover member **70** are disposed outside the pair of protruding pieces **35***a*, **35***a*.

[0097] According to this embodiment, the operating member **50**B further includes the cover member **70** that is disposed on the front side of the operating plate portion **51**, and therefore the design of the operating member **50**B can be changed as appropriate by changing the cover member **70**. In addition, the cover member **70** is provided with the wall portions **75**, **75** that cover the pair of notches **59**B, **59**B formed in the operating plate portion **51**, and therefore, as illustrated in FIG. **18**, when the operating member **50**B is rotated with respect to the base member **30**, the pair of notches **59**B, **59**B can be hidden from view, improving the appearance.

Third Embodiment of Lock Device for Opening-Closing Body

[0098] FIGS. **19** to **21** illustrate a third embodiment of the lock device for the opening-closing body according to the present invention. Parts that are substantially the same as those in the previous embodiment are given the same reference signs, and their descriptions are omitted.

[0099] In a lock device **10**C of an opening-closing body (hereinafter simply referred to as a “lock device **10**C”) of this embodiment, a shaft portion **61**C is separate from the base member **30**. Furthermore, an operating member **50**C has a cover member **70** arranged on the front side of the operating plate portion **51**, as in the second embodiment. Furthermore, a pair of plate-shaped pieces **57***c*, **57***c* protrude from both circumferentially opposing sides of the transmission wall portion **57** of the operating member **50**C. The pair of plate-shaped pieces **57***c*, **57***c* also form the transmission wall portion **57**. Furthermore, the pair of plate-shaped pieces **57***c*, **57***c* are formed with shaft insertion holes **57***d*, **57***d* having a circular hole shape.

[0100] Then, as illustrated in FIG. **20**, the pair of plate-shaped pieces **57***c*, **57***c* of the operating member **50**C are placed inside the pair of protruding pieces **35***a*, **35***a* of the base member **30**, and the shaft portion **61**C is inserted into a shaft support portion **37** placed on each protruding piece **35***a* and the shaft insertion hole **57***d* formed in each plate-shaped pieces **57***c*. Thus, the operating member **50**C is rotatably supported with respect to the base member **30** via the shaft portion **61**C, the shaft support portion **37**, and the shaft insertion hole **57***d*.

[0101] In this embodiment, when the operating portion **72** is rotated to the maximum extent with respect to the base member **30**, the operating load F**1**′ is input from the stopper receiving portion **41** to the stopper portion **63**, and the operating load F**1**′ is transmitted from the stopper portion **63** to the transmission wall portion **57**. Then, the operating load F**1**′ is transmitted through the semicircular portion **57***a* of the transmission wall portion **57** (see the arrow F**2**′ in FIG. **21**) and applied to the pair of plate-shaped pieces **57***c*, **57***c*, and as illustrated by the arrow F**3** in FIG. **21**, the pair of plate-shaped pieces **57***c*, **57***c* attempt to bend and deform in a direction away from each other. In this case, since the pair of restriction wall portions **35**, **35** are arranged to face the pair of plate-shaped pieces **57***c*, **57***c* on the outside of the pair of plate-shaped pieces **57***c*, **57***c*, the deflection deformation of the pair of plate-shaped pieces **57***c*, **57***c* in the radial outward direction of the transmission wall portion **57** can be restricted, and the breaking strength of the operating member **50**C can be increased.

Fourth Embodiment of Lock Device for Opening-Closing Body

[0102] FIGS. **22** to **25** illustrate a fourth embodiment of the lock device for the opening-closing body according to the present invention. Parts that are substantially the same as those in the previous embodiment are given the same reference signs, and their descriptions are omitted.

[0103] A lock device **10**D for the opening-closing body (hereinafter simply referred to as a “lock device **10**D”) of the fourth embodiment illustrated in FIG. **25** has a structure similar to that of the second embodiment, but differs in that a slit **62***a* is formed adjacent to a shaft portion **61**D.

[0104] Specifically, as illustrated in FIG. **22**, a pair of shaft portions **61**D are arc-shaped protrusions integrally formed with the transmission wall portion **57**, and the slit **62***a* is formed at a position adjacent to each shaft portion **61**D. Also, as illustrated in FIG. **24**, a pair of deformation restriction wall portions **35**, **35** of the base member **30** are integrally formed with shaft support portions **37**, **37** The slit **62***a* is formed by an arc-shaped portion extending along a peripheral edge of a tip end side of the shaft portion **61**D in the erection direction of the transmission wall portion **57**, and portions extending parallel to each other from the arc-shaped portion toward the base end side in the erection direction of the transmission wall portion **57**. Through this slit **62***a*, an adjacent portion **62** (hereinafter also simply referred to as a “shaft portion adjacent portion **62**”) of the shaft portion **61**D can be bent and deformed inward and outward of the operating member **50**D.

[0105] In addition, an inward deflection restricting portion **77** that restricts the deflection deformation (here, the deflection deformation of the pair of shaft portion adjacent portions **62**, **62** toward the inside of the transmission wall portion **57**) of the pair of shaft portion adjacent portions **62**, **62** toward the inside of the operating member **50**D is arranged inside the pair of shaft portion adjacent portions **62**, **62**. Here, as illustrated in FIG. **23**, the inward deflection restricting portion **77** having a substantially trapezoidal frame shape that is inserted into the transmission wall portion **57** is erected from the back side of the cover member **70**D, and this inward deflection restricting portion **77** is arranged inside the pair of shaft portion adjacent portions **62**, **62**, so that the deflection deformation of the pair of shaft portion adjacent portions **62**, **62** toward the inside of the transmission wall portion **57** is restricted. In addition, the inward deflection restricting portion **77** as described above may be replaced by a key cylinder **67** to restrict the deflection deformation of the pair of shaft portion adjacent portions **62**, **62** toward the inside of the transmission wall portion **57**.

[0106] In addition, a pair of wall portions **75**, **75** of the cover member **70**D are positioned outside the pair of shaft portion adjacent portions **62**, **62**, so as to restrict deflection deformation of the pair of shaft portion adjacent portions **62**, **62** toward the outside of the transmission wall portion **57** (see FIG. **25**).

[0107] In addition to the above-described embodiment, the structure having the slit may be any of the following embodiments (1) to (5).

[0108] In other words, assuming that the shaft portion is protrusion-shaped and the shaft support portion is groove-shaped (similar to the above structure), the structure may be such that (1) a slit is formed at each of a position adjacent to the shaft portion and a position adjacent to the shaft support portion, so that both the slit adjacent portion of the shaft portion and the slit adjacent portion of the shaft support portion are capable of deflection deformation, or (2) no slit is formed at a position adjacent to the shaft portion, while a slit is formed at a position adjacent to the shaft support portion, so that only the slit adjacent portion of the shaft support portion is capable of deflection deformation
[0109] In addition, assuming that the shaft portion is groove-shaped and the shaft support portion is protrusion-shaped, the structure may be such that (3) a slit is formed at a position adjacent to the shaft portion, while no slit is formed at a position adjacent to the shaft support portion, and only the slit adjacent portion of the shaft portion is capable of deflection deformation, (4) a slit is formed at each of a position adjacent to the shaft portion and a position adjacent to the shaft support portion, and both the slit adjacent portion of the shaft portion and the slit adjacent portion of the shaft support portion are capable of deflection deformation, or (5) no slit is formed at a position adjacent to the shaft portion, while a slit is formed at a position adjacent to the shaft support portion, and only the slit adjacent portion of the shaft support portion is capable of deflection deformation.
[0110] In addition, in this embodiment, the shaft portion **61**D is integrally formed with the operating load transmission wall portion **57**, and the shaft support portions **37** are integrally formed with the pair of deformation restriction wall portions **35**, **35**, but the shaft support portion may be integrally formed with the operating load transmission wall portion, and the shaft portions may be integrally formed with the pair of deformation restriction wall portions.
[0111] In addition, when the slit adjacent portion of the shaft portion or the slit adjacent portion of the shaft support portion can be bent and deformed inward or outward of the operating member or the base member by the above structures (1) to (5), an inward deflection restricting portion or a key cylinder is disposed inside the slit adjacent portion of the shaft portion or the slit adjacent portion of the shaft support portion, and an outward deflection restricting portion is disposed outside the slit adjacent portion of the shaft portion or the slit adjacent portion of the shaft support portion. That is, (a) a structure (this structure is used in the fourth embodiment illustrated in FIGS. **22** to **25**) having only an inward deflection restricting portion or a key cylinder, (b) a structure having only an outward deflection restricting portion, and (c) a structure having both an inward deflection restricting portion or a key cylinder and an outward deflection restricting portion are provided.
[0112] In this embodiment, as illustrated by the arrow F**4** in FIG. **24**, the operating member **50**D is first assembled to the base member **30**. In this case, the pair of transmission wall portions **57**, **57** are inserted inside the pair of restriction wall portions **35**, **35**, and in this case, the tip ends of the pair of shaft portions **61**D, **61**D are pressed by the inner surfaces of the pair of restriction wall portions **35**, **35**, and the pair of shaft portion adjacent portions **62**, **62** are inserted while being bent and deformed inward. When the pair of shaft portions **61**D, **61**D reach the pair of shaft support portions **37**, **37**, the pair of shaft portion adjacent portions **62**, **62** elastically return, and the pair of shaft portions **61**D, **61**D are inserted into the pair of shaft support portions **37**, **37**, and the operating member **50**D is assembled to the base member **30**.
[0113] Thereafter, as illustrated by the arrow F**5** in FIG. **24**, the cover member **70**D is attached to the operating member **50**D. That is, the tip end **52**B of the operating plate portion **51** is inserted into and engaged with the lock frame portion **72***a* of the cover plate portion **71**, and the lock protrusions **73***a*, **73***a* of the cover plate portion **71** are engaged with both sides in the width direction of the base end **53** of the operating plate portion **51**, thereby attaching the cover member **70**D to the front side of the operating plate portion **51**. In this case, as illustrated in FIG. **25**, the inward deflection restricting portion **77** provided on the cover member **70**D is inserted and positioned inside the pair of shaft portion adjacent portions **62**, **62**, and the deflection deformation of the pair of shaft portion adjacent portions **62**, **62** inward of the operating member **50**D, that is, inward of the transmission

wall portion **57**, is restricted.
[0114] As described above, in this embodiment, the operating load transmission wall portion and/or the pair of deformation restriction wall portions have slits formed at positions adjacent to the shaft portion and/or the shaft support portion, and the adjacent portion of the shaft portion and/or the adjacent portion of the shaft support portion can be bent and deformed inward and outward of the operating member or base member through the slits (here, the slit **62***a* is formed at a position adjacent to the shaft portion **61**D, and the shaft portion adjacent portion **62** can be bent and deformed inward and outward of the transmission wall portion **57** of the operating member **50**D through the slit **62***a*), thereby improving the workability of assembling of the operating member **50** to the base member **30**.
[0115] In addition, there is an inward deflection restricting portion or key cylinder that is arranged inside the adjacent portion of the shaft portion or the shaft support portion and that restricts the inward deflection deformation of the operating load transmission wall portion, and/or an outward deflection restricting portion that is arranged outside the adjacent portion of the shaft support portion or the shaft portion and that restricts the outward deflection deformation of the pair of deformation restriction wall portions. Here, the inward deflection restricting portion **77** is arranged inside the pair of shaft portion adjacent portions **62**, **62**. As a result, after the operating member **50**D is assembled to the base member **30**, it is possible to make it difficult for the operating member **50**D to come off from the base member **30**. In other words, the deflection deformation of the slit adjacent portion of the shaft portion and the slit adjacent portion of the shaft support portion formed through the slits can be restricted by the inward deflection restricting portion or the key cylinder, or the outward deflection restricting portion, thereby making it difficult for the operating member to come off from the base member.
[0116] The present invention is not limited to the above-described embodiments, and various modified embodiments are possible within the scope of the gist of the present invention, and such embodiments are also included in the scope of the present invention.
REFERENCE SIGNS LIST
[0117] **1** fixed body [0118] **1***a* opening portion [0119] **1***b*, **1***b* lock portion [0120] **2** opening-closing body [0121] **10**, **10**B, **10**C, **10**D lock device for opening-closing body (lock device) [0122] **20** lock member [0123] **25** rotation member [0124] **27** torsion spring [0125] **30** base member [0126] **35** deformation restriction wall portion [0127] **37** shaft support portion [0128] **41** stopper receiving portion [0129] **50**, **50**A, **50**B, **50**C, **50**D operating member [0130] **51** operating plate portion [0131] **52** operating portion [0132] **57**, **68** operating load transmission wall portion (transmission wall portion) [0133] **59**B, **59**B notch [0134] **61**, **61**C, **61**D shaft portion [0135] **63** stopper portion [0136] **67** key cylinder [0137] **70**,**70**D cover member [0138] **72** operating portion [0139] **77** inward deflection restricting portion

Claims

1. A lock device for an opening-closing body that is configured to be openably and closably attached to an opening portion of a fixed body, comprising: a lock portion provided on one of the fixed body and the opening-closing body; a lock member that is disposed on another of the fixed body and the opening-closing body and that is configured to engage with and disengage from the lock portion; a base member that is fixed to the other of the fixed body and the opening-closing body; and an operating member that is rotatably supported on the base member via a rotation portion and that is configured to operate the lock member, wherein the rotation portion is formed by shaft portions provided on one of the operating member and the base member, and shaft support portions provided on another of the operating member and the base member to rotatably support the shaft portions, the operating member has an operating plate portion extending in a plate shape, an operating portion provided at one end of the operating plate portion in an extension direction, an operating load transmission wall portion protruding from a back side of the operating plate portion, the shaft portions disposed so as to protrude from both sides of the operating load transmission wall portion toward the outside of the operating member or the shaft support portions disposed on the operating load transmission wall portion, and a stopper portion connected to the operating load transmission wall portion on a side of the operating portion with respect to the shaft portions or the shaft support portions, and the base member has a pair of deformation restriction wall portions disposed outside portions of the operating load transmission wall portion where the shaft portions or the shaft support portions are provided, the shaft support portions disposed on the pair of deformation restriction wall portions or the shaft portions disposed to protrude from the pair of deformation restriction wall portions toward the operating load transmission wall portion, and a stopper receiving portion that is capable of coming into contact with the stopper portion and that is configured to restrict rotation of the operating member.

2. The lock device for the opening-closing body according to claim 1, wherein the operating load transmission wall portion is cylindrical shape in which a key cylinder can be inserted.

3. The lock device for the opening-closing body according to claim 1, wherein a pair of notches into which the pair of deformation restriction wall portions are inserted, are formed on both sides of the operating plate portion along the extension direction, and the operating member further has a cover member arranged on a front side of the operating plate portion, and the cover member is provided with a wall portion that cover the pair of notches.

4. The lock device for the opening-closing body according to claim 1, wherein the shaft portions or the shaft support portions are integrally formed with the operating load transmission wall portion, the shaft support portions or the shaft portions are integrally formed with the pair of deformation restriction wall portions, a slit is formed in the operating load transmission wall portion and/or the pair of deformation restriction wall portions at each of positions adjacent to the shaft portions and/or the shaft support portions, and each of adjacent portions of the shaft portions and/or each of adjacent portions of the shaft support portions can be bent and deformed inward and outward of the operating member or the base member via the slit, and an inward deflection restricting portion or a key cylinder that is arranged inside adjacent portions of the shaft portions or the shaft support portions and that is configured to restrict inward deflection deformation of the operating load transmission wall portion and/or an outward deflection restricting portion that is arranged outside the adjacent portions of the shaft support portions or the shaft portions and that is configured to restrict outward deflection deformation of the pair of deformation restriction wall portions are provided.